

COLUMBIA, CA, USA

WMO: 720615

Lat: 38.033N

Lon: 120.417W

Elev: 646

StdP: 93.80

Time Zone: -8.00 (NAP)

Period: 09-19

WBAN: 00206

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-2.9	-2.3	-9.1	1.9	8.5	-7.4	2.2	10.5	5.6	8.3	4.9	8.9	0.1	50	0.446

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	37.9	18.4	37.1	18.0	35.0	17.3	20.0	33.9	19.2	33.0	18.5	32.3	2.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.2	11.6	22.6	14.1	10.8	21.8	13.6	10.5	21.4	59.8	33.8	57.1	32.4	54.5	32.2	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.9	4.4	3.6	-5.7	40.1	1.8	1.5	-7.0	41.2	-8.0	42.0	-9.0	42.8	-10.3	43.9		
			(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-6.5	20.6	1.5	1.1	-7.6	21.4	-8.5	22.1	-9.4	22.7	-10.5	23.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	8.0	8.2	10.0	12.4	15.8	21.3	24.7	24.3	21.5	15.8	10.1	6.7
	DBStd	7.26	3.03	3.31	3.52	3.92	3.98	4.21	2.78	2.56	3.87	3.52	3.66	3.23
	HDD10.0	363	76	67	44	21	3	0	0	0	0	2	43	108
	HDD18.3	1882	321	283	257	182	97	19	0	1	17	96	247	360
	CDD10.0	2169	13	17	45	92	184	338	457	445	346	180	46	7
	CDD18.3	646	0	0	0	3	21	107	198	187	113	16	0	0
	CDH23.3	10632	0	1	17	124	498	1881	3168	2829	1711	380	23	0
CDH26.7	5854	0	0	2	25	169	1023	1928	1662	916	128	1	0	
Wind	WSAvg	1.1	0.7	1.1	1.2	1.3	1.3	1.3	1.3	1.2	1.0	0.9	0.7	0.8
Precipitation	PrecAvg	1155	225	189	149	95	65	19	3	3	9	65	105	212
	PrecMax	1790	556	455	491	295	165	52	42	31	48	211	241	498
	PrecMin	630	17	9	7	9	1	1	0	0	0	1	10	24
	PrecStd	338	161	120	130	76	49	19	9	6	13	61	69	138
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.4	22.5	26.0	29.1	32.9	39.1	39.9	39.1	37.6	32.9	26.0	19.1
		MCWB	10.3	12.8	14.2	16.7	17.1	19.9	18.6	18.4	17.6	15.8	12.9	9.3
	2%	DB	18.9	20.2	22.5	27.2	30.2	37.4	37.9	37.5	35.2	30.0	23.7	17.3
		MCWB	9.5	11.8	13.6	15.9	16.2	18.9	18.3	17.8	16.4	15.0	12.6	9.9
	5%	DB	17.4	18.0	20.2	24.1	28.0	34.8	37.1	35.2	33.9	27.5	21.3	15.0
		MCWB	9.4	11.1	12.2	14.2	15.4	17.8	18.2	17.2	16.4	14.0	12.0	9.3
	10%	DB	15.0	16.0	18.0	22.2	26.3	32.5	35.0	33.9	32.4	25.0	17.8	12.9
		MCWB	9.0	10.0	11.4	13.7	14.3	16.8	17.6	16.8	16.2	13.3	10.8	8.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	14.7	15.5	18.1	18.6	21.3	20.8	20.0	19.7	17.8	16.0	14.0
		MCDB	16.9	18.6	23.3	27.3	30.8	36.0	35.8	34.9	32.6	26.5	22.4	15.8
	2%	WB	12.0	13.2	14.2	16.3	17.2	20.2	19.9	19.1	18.2	16.8	14.1	12.5
		MCDB	14.8	17.8	21.0	25.1	28.5	33.7	34.0	34.3	31.9	26.5	21.4	13.6
	5%	WB	10.9	12.0	13.1	15.0	16.2	19.1	19.1	18.4	17.4	15.5	12.8	11.1
		MCDB	14.5	16.6	19.0	23.2	26.3	31.9	33.4	33.5	31.3	24.8	18.7	12.9
	10%	WB	9.8	10.6	12.0	13.9	15.3	18.0	18.3	17.7	16.6	14.3	11.4	9.8
		MCDB	13.6	15.0	16.9	20.8	25.1	30.1	32.8	32.3	30.5	22.1	16.2	12.3
Mean Daily Temperature Range	5% DB	MDBR	12.7	12.5	12.8	14.6	15.8	18.6	19.5	19.1	18.3	16.0	13.5	11.5
		MCDBR	16.2	16.6	16.8	18.2	19.3	21.0	20.8	20.3	20.4	19.1	17.6	14.4
		MCWBR	8.8	9.2	8.9	8.9	7.7	7.0	6.4	6.7	7.3	7.7	8.7	8.3
	5% WB	MCDBR	12.0	14.7	14.9	17.1	18.0	19.3	19.0	18.9	18.6	16.5	15.4	9.9
		MCWBR	7.7	8.9	8.4	8.9	7.6	6.7	6.1	6.3	6.9	7.2	8.7	7.4
Clear-Sky Solar Irradiance	taub	0.320	0.318	0.341	0.336	0.348	0.340	0.350	0.332	0.335	0.318	0.314	0.298	
	taud	2.427	2.460	2.388	2.432	2.422	2.459	2.451	2.485	2.483	2.530	2.491	2.520	
	Ebn at Noon	858	914	920	943	931	934	923	935	916	903	861	864	
	Edh at Noon	85	93	110	112	115	111	111	105	99	86	79	72	
All-Sky Solar Radiation	RadAvg	2.32	3.27	4.65	6.05	7.33	8.35	8.21	7.43	6.06	4.35	2.85	2.07	
	RadStd	0.36	0.43	0.52	0.56	0.49	0.36	0.29	0.23	0.22	0.29	0.20	0.33	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (46 neighbors)	+0.41	N/A	-1.10	N/A	N/A	N/A	N/A	-116	+139	+42

Nomenclature: See separate page